CS383 · SOFTWARE ENGINEERING

# Midterm 471 Practice

20 marks

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## 01 Questions

Q1 MCQ · 2 PTS

Multiple choice questions

**1. "Software as a Service (SaaS)" slides list one benefit as pay-as-you-go and accessibility, and one drawback as:**

- ☐ Guaranteed performance
- ☐ Full developer control
- ☐ Security/trust vulnerabilities and internet dependence
- ☐ Free on-prem hardware

**2. According to "Application types," what determines which engineering methods/tools to use?**

- ☐ Project manager's personal preference
- ☐ Application type, customer requirements, team background, and tools
- ☐ Software cost alone
- ☐ All of the above

**3. A key weakness of the Waterfall model in the slides is:**

- ☐ Low documentation
- ☐ Iterative
- ☐ Lack of any structure
- ☐ Inflexible to changing requirements and late problem discovery

**4. To cope with change, there are two approaches**

- ☐ Code-and-fix & Big-bang integration
- ☐ Change prevention & big-bang integration
- ☐ Prototyping & incremental delivery
- ☐ Waterfall with extended design phase & late verification

**Q2** USER / SYSTEM · FR / NFR · 4 PTS

For each statement below, classify (i) User vs. System requirement and (ii) Functional (FR) vs. Non-functional (NFR).

| Requirement | User vs. System | FR vs. NFR |
|---|---|---|
| "The system should be easy to use for first-year students." | | |
| "The application shall allow password reset using a verified email." | | |
| "The report generation shall be fast." | | |
| "The interface will display student GPA to two decimal places." | | |

**Q3** SPRINT CALCULATIONS · 4 PTS

Team of 6 people; workday = 8h. Last 4 sprints completed story points: 22, 28, 26, 24.

**1. Compute the sprint velocity (use the average sprint story points). (1pts)**

**2. Compute focus factor (SprintVelocity ÷ (people × hours per day)). (1pts)**

**3. For a 3-week sprint (Mon–Fri, weekends off), compute effective team capacity (hours) using the focus factor. Show formulas and round sensibly. (1pts)**

**4. If the next sprint backlog is 30 story points, state whether the plan is feasible and why. (1pts)**

Feasibility: ☐ Feasible or ☐ Prudent Justification:

**Q4.1** PROCESS MODEL · 4 PTS

For each project, choose one development process model: Agile, Plan-Driven, or Component-Based.

**Project 1 — Smart-City Sensor Hub**

• Stakeholder Presence: Moderate; city IT holds bi-weekly reviews; end-users (operators) rarely available.

• Requirement Volatility: Medium; new sensor types and vendors appear mid-project, but core data collection goals are stable.

• Project Size & Team: 20 engineers across 3 vendors; distributed.

• Architecture Constraints: Must integrate **COTS gateways** and vendor SDKs; strict uptime (99.9%).

• Compliance: Must pass basic cybersecurity hardening but no heavy certification.

Choose: ☐ Agile ☐ Plan-Driven ☐ Component-Based

**Q4.2** PROJECT 2

**Project 2 — Banking Regulatory Reporting**

• Stakeholder Presence: Limited; auditors and compliance office set requirements up front and review only at stage gates.

• Requirement Volatility: Low; templates and KPIs fixed by regulation for the next 3 years.

• Project Size & Team: 30 people, multi-department; formal PMO.

• Documentation & Traceability: Detailed SRS and SDD, audit trails, sign-offs required.

• Risk Profile: High penalty for defects; schedule predictable.

Choose: ☐ Agile ☐ Plan-Driven ☐ Component-Based

Q4.3 PROJECT 3

**Project 3 — Campus Social-Learning MVP**

• Stakeholder Presence: High; product owner and pilot students give **daily** feedback.

• Requirement Volatility: High; features frequently reprioritized to find product-market fit.

• Project Size & Team: 6 developers, co-located; CI and CD in place.

• Time to Market: Need public beta in **4–6 weeks**; success measured by adoption and engagement.

• Architecture: Greenfield; minimal legacy constraints.

Choose: ☐ Agile ☐ Plan-Driven ☐ Component-Based

Q4.4 PROJECT 4

**Project 4 — Medical Device Companion App**

• Stakeholder Presence: Moderate; clinicians available for periodic usability reviews; regulatory team sets constraints.

• Requirement Volatility: Medium; UI iterations expected, but clinical workflows and data formats are fixed by standards.

• Project Size & Team: 15 people; UX + mobile + back-end.

• Compliance: Must align with **IEC 62304** software life-cycle documentation and **ISO 14971** risk management; app is Class II accessory.

• Integration: Uses a certified SDK from the device vendor for data acquisition.

Choose: ☐ Agile ☐ Plan-Driven ☐ Component-Based

**Q5** PRODUCT / ORGANIZATION / EXTERNAL · 6 PTS

For each statement, indicate the requirement type: Product, Organization or External

**1. The API shall respond in ≤ 300 ms (p95) for read requests under 100 concurrent users.**

Choose: ☐ Product ☐ Organization ☐ External

**2. All new services must be built in Java 21 and deployed on the organization's Kubernetes platform.**

Choose: ☐ Product ☐ Organization ☐ External

**3. Patient data must comply with HIPAA and local privacy law X.**

Choose: ☐ Product ☐ Organization ☐ External

**4. The mobile app shall support offline read-only access for 24 hours.**

Choose: ☐ Product ☐ Organization ☐ External

**5. The system shall export reports in CSV and PDF.**

Choose: ☐ Product ☐ Organization ☐ External

**6. Device firmware must interoperate with ISO/IEC 14443 Type A NFC readers.**

Choose: ☐ Product ☐ Organization ☐ External